

B. Tech Degree V Semester Examination, November 2008**ME 504 THERMAL ENGINEERING**

(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A(Answer ALL questions)

(5 x 8 = 40)

- I. (a) With the help of a T-S diagram derive the expression for thermal efficiency in terms of compression ratio for an otto cycle.
 (b) Explain the valve timing diagram for the four stroke cycle spark ignition engine.
 (c) Describe the working of a Wankel engine.
 (d) Make a comparison between open and divided combustion chamber in IC engines.
 (e) Explain the effect of friction on the performance of a steam nozzle.
 (f) Explain the difference between the impulse and reaction turbine.
 (g) Explain the ideal regenerative Feed heating cycle.
 (h) With the help of neat sketch explain the working of axial flow air compressor.

PART B

(4 x 15 = 60)

- II. The compression curve on the pressure-volume diagram of a gas engine working on Ottocycle follows the polytropic law $PV^{1.3} = \text{constant}$. At two points on this curve at $\frac{1}{4}$ th stroke and $\frac{3}{4}$ th stroke, the pressure are measured to be 140kN/m² and 360kN/m² respectively. Determine the compression ratio of the engine. Also calculate the gas consumption per KW hour if relative efficiency is 0.4 and the gas has calorific value of 18840kJ/m³. (15)

OR

- III. An ideal diesel engine operates on 1Kg of air with suction at 1 Kg/cm² and 37°C. The pressure at the end of the compression is 33 kgf/cm² and the cut-off takes place at 6% of the stroke from the inner dead centre portion. Treating the working substance as air, determine
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|-------|------------------------|------|----------------------|
| (i) | Compression ratio | (ii) | Percentage clearance |
| (iii) | The heat supplied | (iv) | The heat rejected |
| (v) | The thermal efficiency | | |
- (15)

- IV. (a) Explain with the help of neat sketches ignition system for a spark ignition engine. (8)
 (b) Explain various methods of cooling internal combustion engines. (7)

OR

- V. (a) Explain the phenomena of detonation in spark ignition engine. What are the main factors which affect the detonation? (8)
 (b) Explain the diesel knock. On what factors does it depend. (7)

- VI. An adiabatic steam nozzle is to be designed for a discharge rate of 10Kg/s, of steam from 10 bar and 400°C to back pressure of 1 bar. The nozzle efficiency is 0.92 and the frictional loss is assumed to take place in the diverging portion of the nozzle only. Assume a critical pressure ratio of 0.5457. Determine the throat and exit area. (15)

OR

- VII. In a stage of an impulse turbine provided with a single row wheel, the mean diameter of the blade is 1m. It runs at 300 rpm. The steam issues from the nozzle at a velocity of 350m/s and the nozzle angle is 20°. The rotor blades are equiangular. The blade friction factor is 0.86. Determine the power developed if axial thrust on the end bearing of a rotor is 120N. (15)

- VIII. Explain with the help of a temperature – entropy diagram and gas turbine with inter cooling and reheating: (15)

OR

- IX. In a gas turbine plant, the air at 283K and 1 bar is compressed to 4 bar with compression efficiency of 80%. The air is heated in the re-generator and in the combustion chamber till its temperature is raised to 973 K, and during the process the pressure falls by 0.1 bar. The air is then expanded in turbine and passes to the regenerator which has 75% effectiveness and causes a pressure drop of 0.14 bar. If the isentropic efficiency of the turbine is 0.85 determine the thermal efficiency of the plant. (15)

